

RiskLens

AI-Powered Credit Risk Intelligence Platform

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Streamlit · LightGBM · SHAP · Groq LLaMA 3.3

Use Case & Business Value

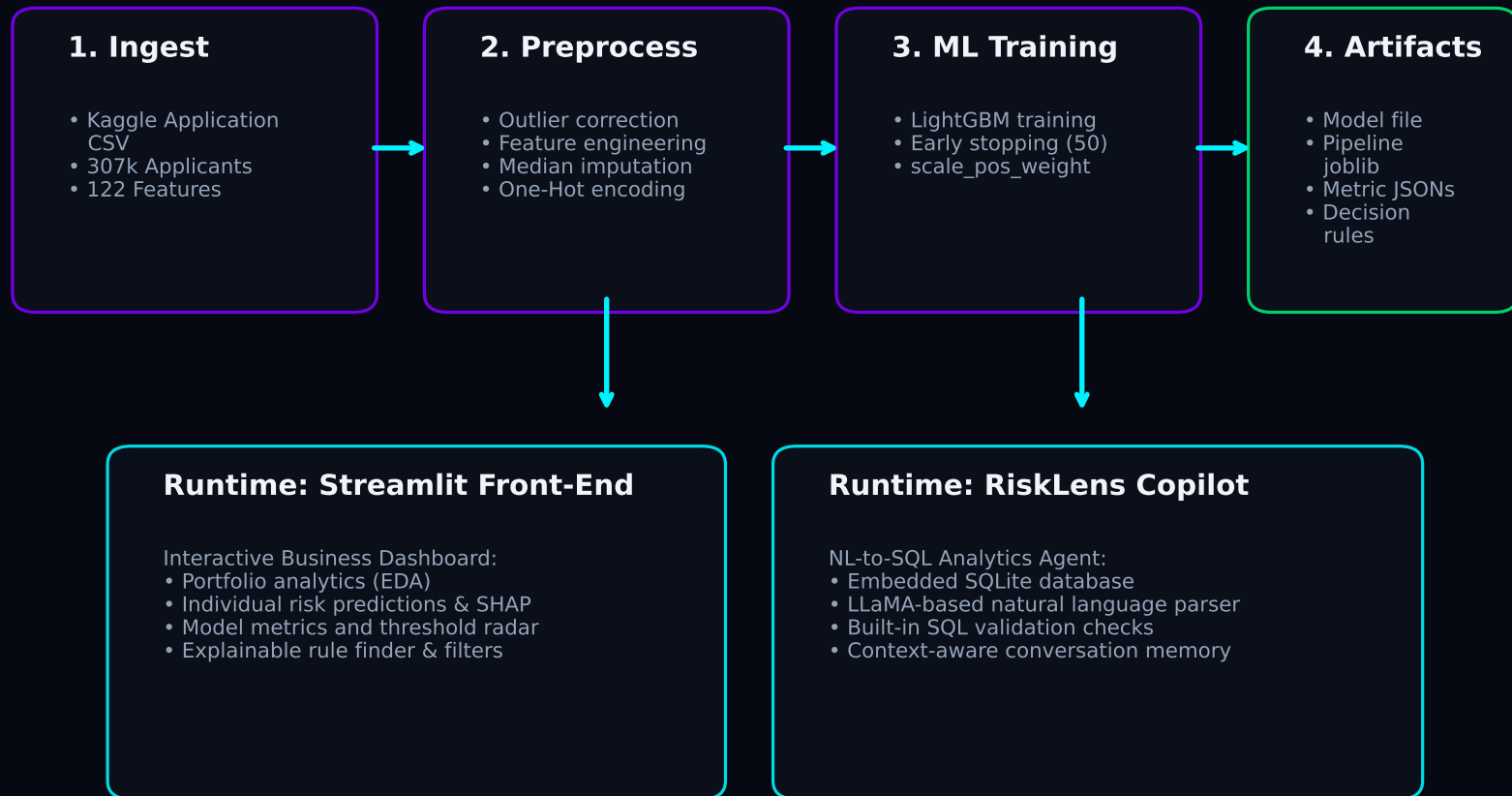
The Credit Risk Challenge

- High Default Losses:
Unpaid loans drive heavy financial exposure.
- Black-Box ML Models:
Accurate models lack explainability, failing regulatory audits (e.g., SR 11-7).
- Complex Data Access:
Risk teams cannot explore portfolio metrics without technical SQL knowledge.

The RiskLens Core Solution

- ✓ Accurate Scoring:
Real-time default risk prediction using optimized threshold tuning.
- ✓ Human-Readable Rules:
Surrogate decision trees convert black-box predictions into transparent rules.
- ✓ Global & Local Explainability:
SHAP integration provides instant feature attribution for compliance.
- ✓ AI Copilot Agent:
Conversational interface translates natural language queries into instant SQL.

Platform System Architecture



Model Performance & Imbalance Strategy

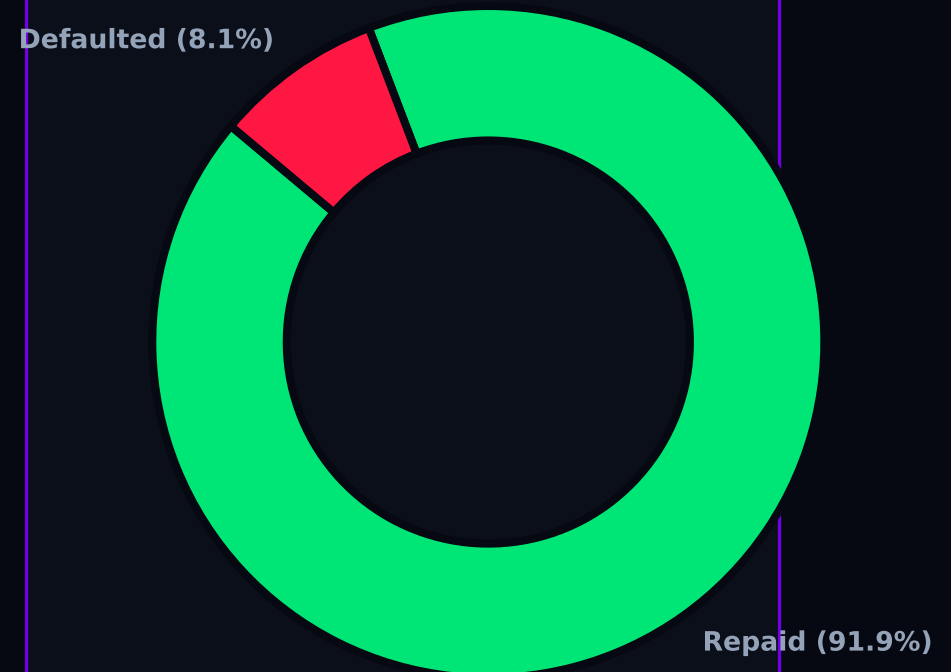
Model Metrics Report

- ROC-AUC Score: 0.7683 (Robust separation)
- PR-AUC Score: 0.2591 (3.2x random baseline)
- F1 Score: 0.3202 (Optimized for minority)
- Calibrated Threshold: 0.6815
- Validation Accuracy: 86.0%

Class Imbalance Strategy

- Skewed Dataset:
~92% repaid, ~8% defaulted. Standard accuracy is deceptive.
- scale_pos_weight Tuning:
Penalizes missed defaults 11.3x more heavily than false positives.
- Custom Classification Threshold:
Set at score ≥ 0.6815 to maximize minority-class F1-score.

Applicant Default Imbalance (307k)



Model Transparency & Rules Engine

SHAP Local & Global Explanations

- Local & Global SHAP Values:
Uses shap.TreeExplainer for exact risk contributions.
- Individual Risk Shift:
Highlights features that increase risk (red) or reduce risk (green) relative to the base average.
- Top Influential Features:
 1. External Credit Scores (EXT_SOURCE_2/3)
 2. Age (AGE_YEARS)
 3. Leverage (CREDIT_INCOME_RATIO)

Surrogate Rule Derivation

- Model Approximation:
Shallow decision tree models the LightGBM decision boundary.
- Key Metrics:
 - 16 human-readable rules
 - 85.73% surrogate fidelity (agreement rate)
- Calibrated Risk Bands:
Low, Medium, High risk bands derived from leaf nodes.
- Sample Decision Rule:
IF Credit Score 2 $\leq$ 0.4305 AND Credit Score 3 $>$ 0.31
AND Employed $>$ 4.5 years
--> RISK BAND: Medium (58.9% Confidence)

RiskLens Copilot — Conversational Analyst

Copilot Agent Core Capabilities

- Natural Language to SQL:
Converts user queries to SQLite code and runs them instantly.
- Contextual Memory:
Remembers previous questions (memory depth k=5).
- Guardrails & Security:
 - Restricts access to read-only SELECT commands.
 - Blocks destructive updates (DROP/DELETE).
 - Caps response payloads to 100 rows.

Verified Query Patterns & Examples

- ✓ Aggregations & Metrics:
 - 'What is the average income of defaulted applicants?'
 - SQL: `SELECT AVG(AMT_INCOME_TOTAL) FROM app WHERE TARGET=1`
- ✓ Grouped Rates:
 - 'Show default rate by education level'
 - SQL: `SELECT education, AVG(TARGET) FROM app GROUP BY education`
- ✓ Filters & Counts:
 - 'Find count of female applicants with >2 children'
 - SQL: `SELECT COUNT(*) FROM app WHERE gender='F' AND children > 2`